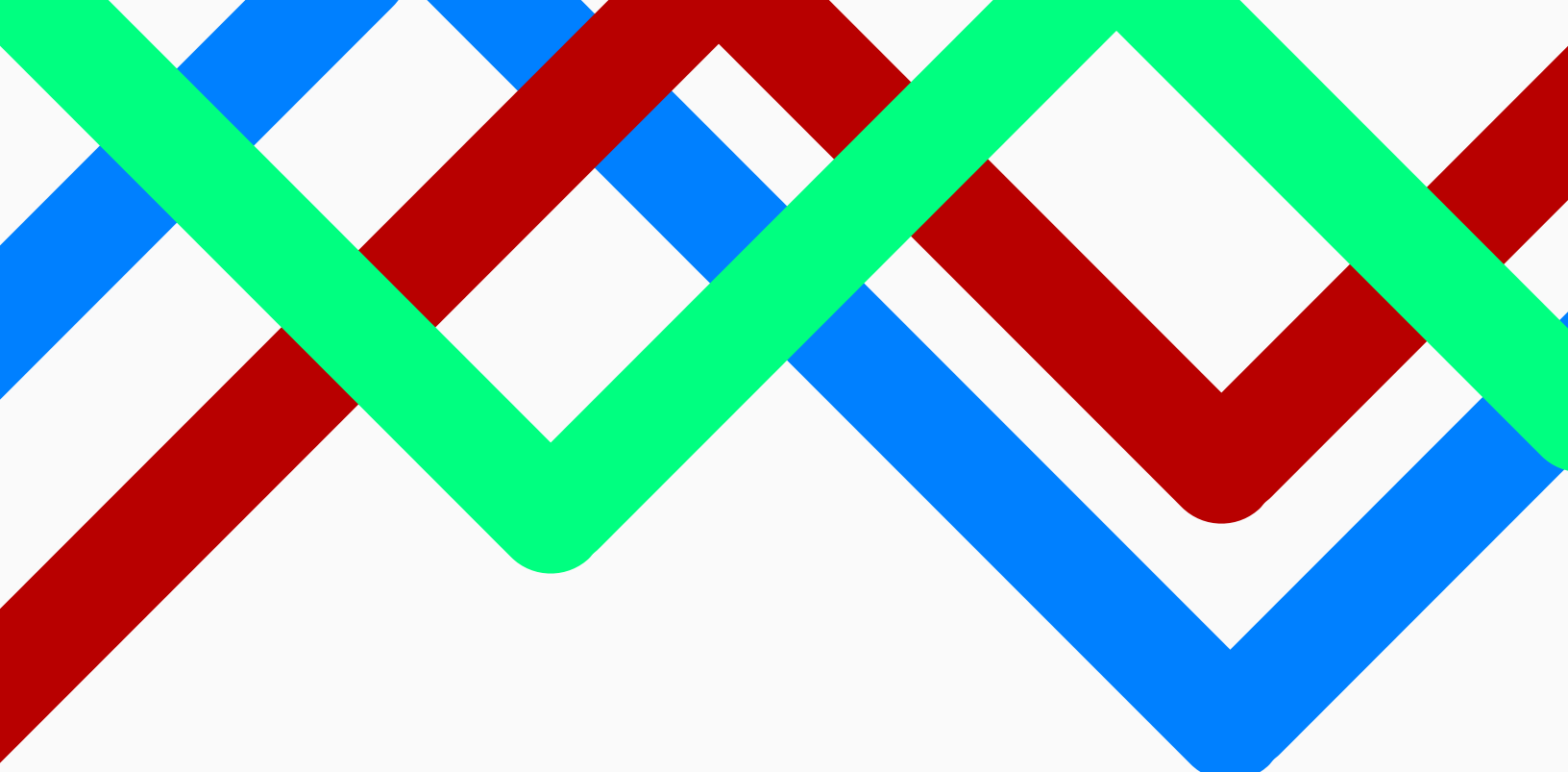




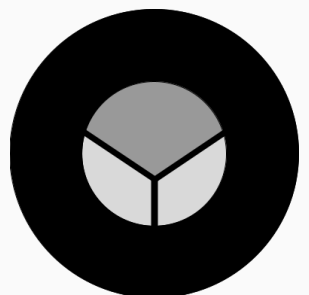
Enzyme Lab Report

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DE Intro to Biology I

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“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



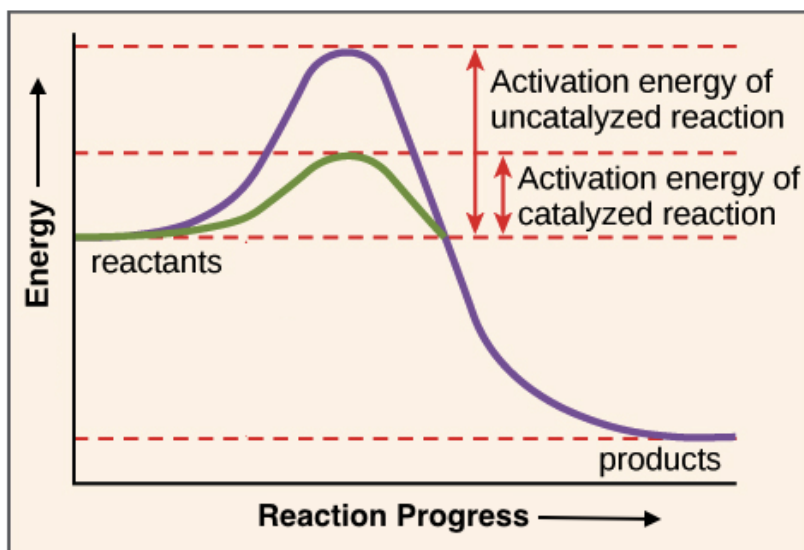
1 Abstract

Amylase, an enzyme found in saliva, breaks down polysaccharides into sugars such as maltose, which breaks down into glucose. This lab used human saliva, saltines, and solutions of various pH levels to determine the correlation between pH and α -amylase efficiency. The data showed no notable correlation between pH levels and glucose levels in each individual solution. However, a second conclusion was drawn; human saliva contains no maltase, and hence, cannot break down polysaccharidic starches into glucose via the enzymes found in human saliva. This impacts the knowledge that saliva cannot break down polysaccharides into glucose.

2 Background / Introduction

Many students recall doing a lab during elementary / middle school biology, whereas the students chew on a cracker for several minutes. The students note that as time goes on, the cracker tastes sweeter and sweeter. This is due to a common enzyme in saliva, called amylase.

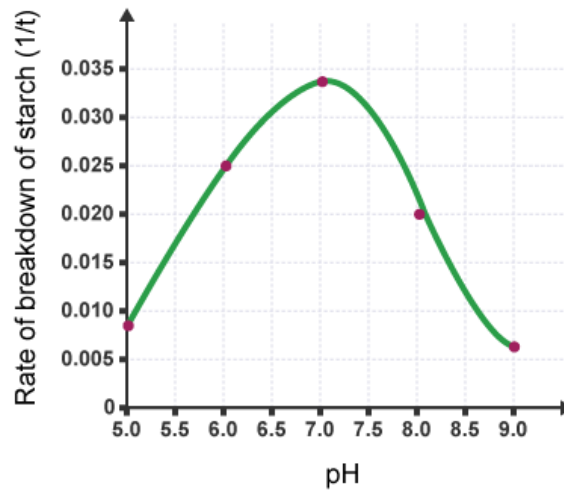
An enzyme is a catalyst that speeds up the rate of a reaction. An enzyme is not destroyed during this reaction and can be used multiple times. Enzymes work by lowering activation energy of a catalyst reaction.



Enzymes have four factors that affect rate of reaction. Each specific enzyme works best at specific temperatures and pH levels. Generally, rising temperature speeds up a reaction, but extreme heat may cause an enzyme to denature. Additionally, each enzyme has an ideal pH range. Extreme pH values may cause enzymes to denature. Denaturing occurs when the enzyme loses its shape due to these stressing factors. This lab explores denaturing of a specific enzyme, amylase, via acidity.

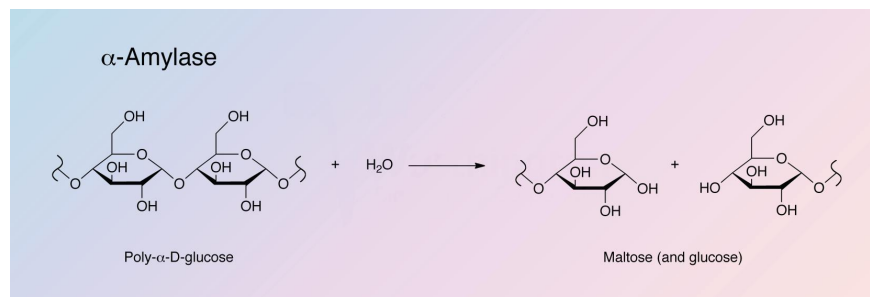
Amylase is an enzyme that helps with the digestion of carbohydrates. It is made in the pancreas and in saliva glands. Amylase is categorized into "3 main classes -- alpha-, beta-, and gamma amylases-- each targeting distinct segments of the carbohydrate molecule. Alpha amylase is present in humans, animals plants and microbes, whereas beta amylase is primarily found in microbes and plants. Gamma amylase, on the other hand, can be found in both animals and plants." (Akinfemiwa, 2023) Amylase also stays stable for 4 days at room temperature, so there is no worry about spoiling of amylase.

α -amylase has an optimal pH of 7, following this curve.



α -amylase splits linear carbohydrate polymers at internal bonds. (Akifemiwa, 2023) It does so via the process of hydrolysis. Considered the reverse of the polymer building process dehydration synthesis, "(hydrolysis) is a chemical process in which a molecule is cleaved into two fragments by the addition of a molecule of water. One fragment of the parent molecule gains a hydrogen ion (H^+), while the other group collects the remaining **hydroxyl** group (OH^-)."

(Parekh, 2016) Amylase completes this by cleaving a starch (a polysaccharide) into "smaller molecules, ultimately yielding maltose, which in turn is cleaved into two glucose molecules." (Gachons, 2016)



Since amylase has an optimal pH of 7, which is the pH of water, it shows that amylase would work the best (pH wise) if in a solution of pH 7. Note, though, that maltose does not break down into glucose naturally. Maltose is "broken down into glucose in the small intestine by the enzyme maltase, facilitating its absorption into the blood stream." (Rogers) Hence, it is not reasonable to assume that maltase is inside of saliva since maltase is found in the small intestine.

3 Question / Problem

How does *amylase*, an enzyme found in saliva, break polysaccharides into simpler carbohydrates at different PH levels?

4 Elements & Hypothesis

4.1 Hypothesis

If the pH levels of three solutions of pH {4, 5.5, 7}, are exposed to the enzyme *amylase* found in saliva and mixed with a polysaccharide containing cracker, then there should be a trend whereas the pH 7 solution will have the highest density of glucose, the pH 5.5 solution should have the medium density of glucose, and the pH 4 solution should have the lowest density of glucose.

4.2 Experimental Elements

- Independent Variable: pH Level of the solution
- Dependent Variable: Glucose concentration produced (in mmol/L)
- Experimental Groups:
 - Group 1: Saltine Cracker, Saliva, pH 4.0 solution
 - Group 2: Saltine Cracker, Saliva, pH 5.5 solution
- Positive Control Group:
 - Saltine Cracker, Saliva, optimal pH solution (pH 7.0)
- Negative Control Group:
 - Saltine Cracker, optimal pH solution (pH 7.0)
- Controls
 - Saliva to Solution ratio
 - Cracker to Solution ratio
 - Time allowed for enzymatic processes
 - Method of glucose measurement

5 Materials / Set up

5.1 Materials

The following materials were gathered:

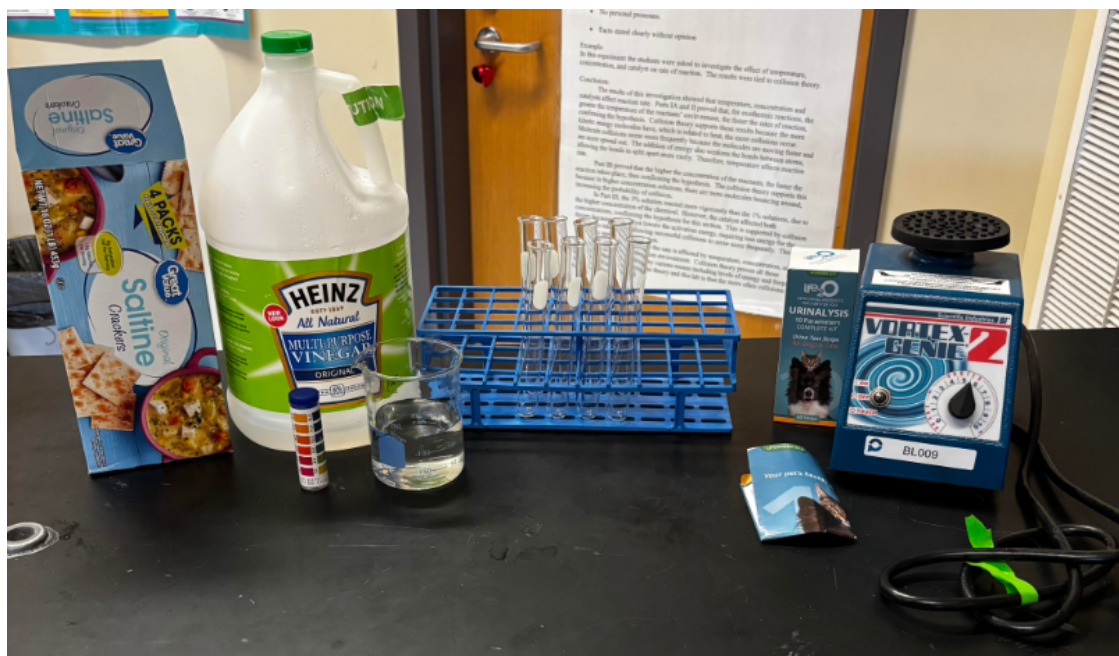
1. 4 saltine cracker pieces (approximately 1 gram each)
2. Roughly 50 mL of Heinz all-natural vinegar
3. pH test strips
4. Roughly 100 mL of distilled water
5. 8 15-25mL test tubes
6. Test tube rack
7. Life2O Pet Urinalysis Test Strips: Dog and Cat Urine Test Strips
8. Genie 2 Vortex Mixer SI-0236
9. Fresh Human Saliva (Roughly 20 mL total)
10. Stopwatch
11. Mortar & Pestle
12. Plastic test tubes
13. Plastic gloves
14. Tape
15. Black Marker
16. 2x 250 mL beaker

5.2 Set Up Procedure

1. All lab materials were collected and thoroughly inspected to ensure cleanliness.
2. Two sets of four test tubes were labeled with {"C","pH=7","pH=5.5","pH=4"} using tape and a marker.

3. The first set of four test tubes were filled with 10 mL of distilled water.
 - The water was filled up in the first beaker with about 100 mL of distilled water.
 - The vinegar was filled up to about the 50 mL mark of the second beaker.
4. Then, according to the test tube labels, the first set of four had 2 mL of vinegar added incrementally (until the pH was correct) and checked via a pH strip.
5. When all four test tubes of the first set were at the pH level indicated by each respective label, the tubes were put away, in a row, back onto the test tube rack
6. All lab members put on their gloves.

An exemplar image of an example setup without pH solutions can be found below:



6 Methods / Procedure

1. Saliva was collected by each individual labmate into their respective plastic test tube. This was to minimize the safety impact of having different contaminants on each test tube.
2. Four crackers (approximately 1 gram each) were crushed by a mortar and pestle into a fine powder.
3. The four crackers were distributed evenly across the **second** set of test tubes. (So now, the second set only has cracker powder in it.)

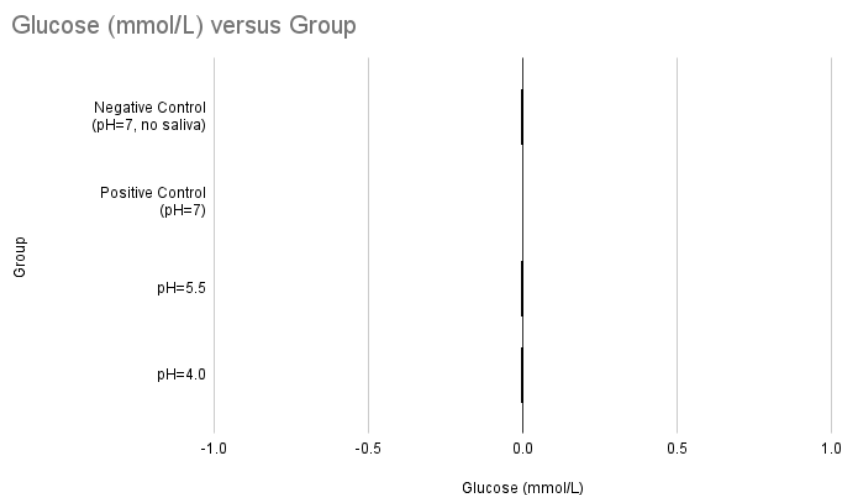
4. The saliva was mixed into a fourth plastic test tube and vortexed via the Genie 2 Vortex Mixer
5. The labmates were careful here; the saliva was then distributed evenly into the **first** set of test tubes. (So now, the first set only has solution + saliva. Only $\frac{3}{4}$ contained saliva because of the negative control.)
6. Carefully, the first set of test tubes was added onto each respective test tube of the second set and began a stopwatch.
7. This is where the rush began. During the 30 seconds of curing (determined via the stopwatch), each test tube was vortexed to ensure that each solution was at least somewhat homogenous. This was a step where caution was taken to ensure that mixture escaped the test tube.
8. After the 30 seconds of curing (determined via the stopwatch) was over, one urineanalysis strip was placed into each test tube to allow for glucose measurement.
9. The stopwatch was reset and began counting to 30 seconds again. After the 30 seconds were over, each urineanalysis strip was carefully removed from each test tube. Then, the urineanalysis strip was compared to the package, and the glucose levels were measured and recorded.

7 Data / Results

7.1 Table

Group	Glucose Measured (mmol/L)
Negative Control (pH=7.0, NO saliva)	0
Positive Control (pH=7.0)	0
pH=5.5	0
pH=4.0	0

7.2 Graph



8 Conclusion / Discussion

The data observed rejects the hypothesis. Only the negative control group and the pH 4.0 experimental group behaved as expected. Since the negative control does not contain saliva, it shouldn't have amylase. This removed sources of possible error from faulty urinalysis strips. However, the positive control and the experimental pH 5.5 group behaved unexpectedly. This is due to faulty research in the very beginning of the lab procedure. Amylase breaks polysaccharides into maltose, which is a disaccharide of two glucoses. Since the lab experiment measured levels of glucose, the urinalysis was looking for the wrong molecule. Since maltose "catalyzes the hydrolysis of the disaccharide maltose to the simple sugar glucose," (The Editors) and is found "synthesised by cells of the mucous membrane lining the intestinal wall," (The Editors) it is unlikely that any maltase is present in saliva. This fact, in juxtaposition with the fact that only maltose is in saliva, makes the probability of glucose in each test group relatively small. This would have been avoided if the lab included maltose indicators *or* if there was maltase present in the solution. Hence, the data observed rejects the hypothesis.

The common misconception that amylase breaks starches into glucose comes from a common elementary school lab. Students are instructed to chew a saltine cracker for an exorbitant amount of time; eventually, the saltine tastes sweet inside the mouth. Since sweetness is associated with sugar, many students falsely correlate glucose with the sweetness produced by saliva.

In the future, the data shows that future labs should either include artificial curing of the enzyme maltase

to break down the maltose molecules into glucose, so that the urineanalysis strips can detect the glucose, or that urineanalysis strips that measure the presence of maltose should be purchased.

9 References

- Academy, K. (n.d.). Enzymes Review. Khan Academy. <https://www.khanacademy.org/science/ap-biology/cellular-energetics/environmental-impacts-on-enzyme-function/a/hs-enzymes-review>
- Akinfemiwa, O. (2023, November 12). Amylase. National Library of medicine. <https://www.ncbi.nlm.nih.gov/books/NBK557738/> Amylase, alpha - Worthington Enzyme Manual. Worthington Biochemical. (n.d.). <https://www.worthington-biochem.com/products/amylase-alpha/manual>
- Brodkey, F. D. (n.d.). Amylase - blood. MedlinePlus. <https://medlineplus.gov/ency/article/003464.htm>
- The Editors of Encyclopædia Britannica (n.d.). Maltase. Encyclopædia Britannica. <https://www.britannica.com/science/maltase>
- Enzyme. Genome.gov. (n.d.). <https://www.genome.gov/genetics-glossary/Enzyme>
- News, B. (2025, April 24). The effect of ph on the reaction rate of amylase - what happens in cells and what do cells need? - OCR gateway - GCSE combined science revision - OCR gateway - BBC bitesize. BBC News. <https://www.bbc.co.uk/bitesize/guides/z9jrng8/revision/4>
- Parekh, V. J. (n.d.). Engineering Fundamentals of Biotechnology. Science Direct. <https://www.sciencedirect.com/topics/immunology-and-microbiology/hydrolysis>
- Peyrot des Gachons, C., & Breslin, P. A. S. (2016, October). Salivary amylase: Digestion and metabolic syndrome. Current diabetes reports. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6825871/>
- Rogers, K. (n.d.). Maltose. Encyclopædia Britannica. <https://www.britannica.com/science/maltose>